

On the Dynamical Isomorphism of Periodic Messengers: The R_{yy} Temporal Manifold Extension 8

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Abstract

The generalized deployment of the Q-LINK architecture requires flexibility in both finite-depth optimization dynamics and hardware-native interaction bases. This note investigates Extension 8: the combination of a p -periodic residual messenger channel utilizing an R_{yy} collection basis. We formally evaluate the iterated Lie brackets under temporal sparsity, proving that the p -periodic R_{yy} architecture strictly preserves the algebraic isomorphism to the dense R_{xx} baseline. We conclude that this combination mathematically guarantees the exact asymptotic variance bound $\mathcal{R}(n) = \frac{n+1}{n}$, while providing a highly elastic framework for navigating finite-depth trajectory variance on YY -native hardware topologies.

1 The Periodic R_{yy} Generator Set

Let the periodic R_{yy} architecture be defined by applying the data-register operations U_{opr} at every layer, and the residual mechanism ($U_{coll}^{(Y)}$ and U_{dist}) strictly every p layers.

The available Lie generator set during the p -th layer is the union of three components:

- $\mathcal{G}_{opr} = \{I_m \otimes A\}$ for $A \in \mathfrak{su}(2^n)$.
- $\mathcal{G}_{coll}^{(Y)} = \{Y_m \otimes Y_i\}$ from the R_{yy} collection interactions.
- $\mathcal{G}_{dist} = \{Z_m \otimes X_i\}$ from the messenger-controlled CNOT distribution.

2 DLA Isomorphism Under Temporal Sparsity

As established in prior analysis, the Dynamical Lie Algebra (DLA) is the closure of the generators under nested commutators, $\mathfrak{g} = \langle \mathcal{G} \rangle_{Lie}$. The continuous algebraic closure is invariant under the discrete periodic application of the generators.

We evaluate the commutator between the collection and distribution layers:

$$[Y_m \otimes Y_i, Z_m \otimes X_i] = 2(X_m \otimes Z_i) \quad (1)$$

Because the Y_m and Z_m components do not commute, the Lie bracket successfully breaks out of the Y -eigenbasis, generating the orthogonal X_m operator space.

By iterating commutators between $\{I_m \otimes A, Y_m \otimes B, Z_m \otimes C, X_m \otimes D\}$, the algebra strictly closes at the full special unitary group. Therefore, for any period $p \geq 1$:

$$\mathfrak{g}_p^{(Y)} = \langle \mathcal{G}_{opr} \cup \mathcal{G}_{coll}^{(Y)} \cup \mathcal{G}_{dist} \rangle_{Lie} \cong \mathfrak{su}(2^{n+1}) \quad (2)$$

Theorem of Isomorphism: The temporal sparsity of the residual channel ($p > 1$) does not fracture the algebraic expansion of the R_{yy} interaction basis. The resulting continuous manifold is identical to the dense R_{xx} baseline.

3 Trace Invariance and Asymptotic Bounds

Because the periodic R_{yy} architecture perfectly saturates the extended $U(2^{n+1})$ Haar measure, the integration volume utilized in the Master Formula is identical to the unablated baseline.

Assuming the standard global average cost operator $O = I - \frac{1}{n} \sum Z_i$ (which acts trivially on the messenger), the traceless Hilbert-Schmidt norm remains:

$$\text{Tr}(\tilde{V}^2) = \frac{2N}{n} \quad (3)$$

Taking the ratio against a Vanilla circuit on $n+1$ total qubits at depth saturation, the asymptotic relative variance bound is strictly preserved:

$$\mathcal{R}_p^{(Y)}(n) = \frac{n+1}{n} \quad (4)$$

4 Conclusion and Physical Implications

Extension 8 is mathematically sound and structurally robust. While it does not alter the asymptotic geometric bounds of the network, the R_{yy} periodic manifold provides two distinct engineering advantages. First, the period p acts as a temporal throttle on operator spreading, delaying exact t -design convergence and widening the operable finite-depth trajectory variance window. Second, the rigorous proof of isomorphism allows experimentalists to port the Q-LINK architecture directly to hardware topologies with native YY -couplings without sacrificing the barren plateau mitigation guarantees derived for the XX -coupled baseline.